



Preparation handout for

BPM2 DM Small Group

HPWP 04: Selecting & Appraising Data for Self-Directed Learning

Contemporary Data Selection

In BPM1, it was taught that developing skills in self-directed learning (SDL) and metacognition is essential for medical school and future clinical practice. In contemporary society information is abundant, with over 45,000 academic journals covering 50 million academic articles and over 2 million more articles added per year, which in turn leads to a doubling time of medical knowledge of less than 73 days. This data deluge makes these skills more important than ever.

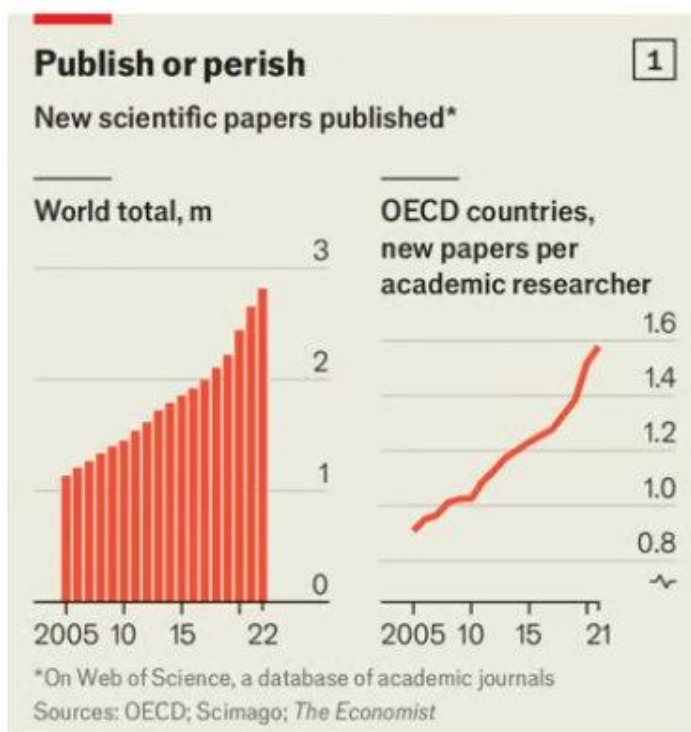


CHART: THE ECONOMIST

From *The Economist*, 2024

Beyond navigating an overwhelming flow of information, medical practitioners may also encounter research findings that do not appear reliable or seem implausible. Since clinical decisions often rely on the latest research, the ability to critically assess studies is paramount and SDL skills enable life-long learners to seek out original research, evaluate study design, and question the robustness of findings.

Appraisal Methods

To effectively traverse these challenges, medical professionals need to develop the ability to critically appraise medical literature. This appraisal should cover all sections of any published medical journal article, including the Introduction, Methods, Results, and Discussion.

The Introduction should establish valid standards and criteria, as well as ensure that novel data or designs can be explained by current scientific principles. Within the Methods, a rational investigative design should be clear and measurable with careful consideration of potential confounds and inclusion of appropriate comparator groups. In this section, it is also important to assess potential conflicts of interest that could influence experimental design and compromise the integrity of research. For instance, a 2015 study examining over 300 randomized controlled trials of medications, biologics and devices found that industry-sponsored trials systematically yielded favorable results for the sponsors.

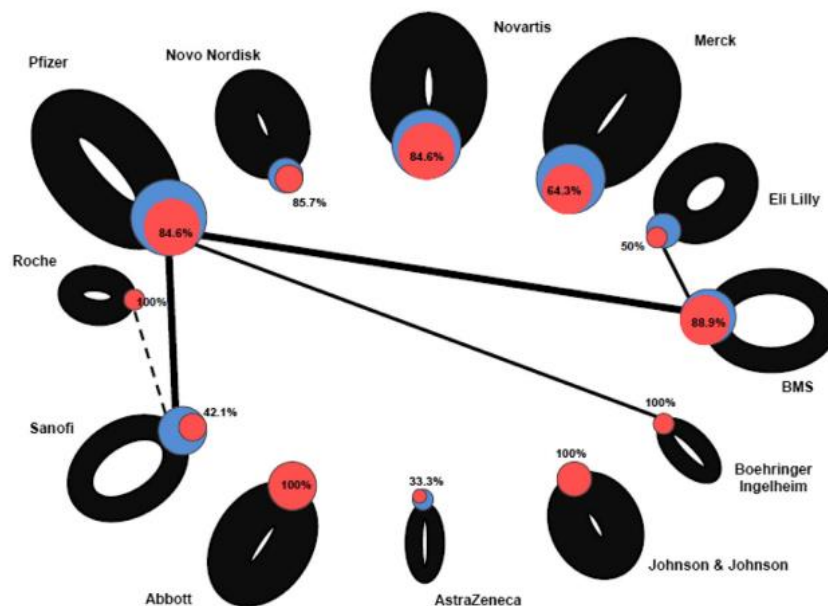


Fig. 2. Network of cosponsorship for the 12 most prolific companies. Each company is shown by a node whose diameter is proportional to the number of trials sponsored. Lines represent cosponsorship between companies, with thickness proportional to the number of trials cosponsored. Dashed lines refer to comparison with only one cosponsored trial. The thickness of the autoloops is proportional to the number of trials where the respective company is the unique sponsor. Red circles into each node represent the proportion of trials sponsored (or cosponsored) by each company that showed favorable results to the specific company (corresponding percentages are listed next to each node). BMS, Bristol-Myers Squibb.

From Flacco *et al.*, 2015

In Results, data should be transparently reported and presented findings should be plausible with respect to the study design and understood scientific mechanisms. Discussion and interpretation of the data also need to account for cognitive biases that may influence how evidence is presented or perceived. Finally, conclusions offered must be substantiated so that the study's findings can be given appropriate contextualization and lead to accurate applications of the findings in clinical practice.

While not exhaustive, these minimum steps will allow for critical engagement with research and start to foster evidence-based decision-making. Building on the same critical appraisal skills, the Fresno Test of Competence in Evidence-Based Medicine provides a structured framework to guide student and professional learners to evaluate research with greater depth and precision. The framework explicitly promotes formulating clinical questions, searching for relevant evidence, critically appraising research, and applying findings to patient care. A shortened version of the standardized rubric is shown below:

Short answer question	Ideal answer in rubric	Lesser answer in rubric	Poor answer in rubric
1) Form a PICO question			
1a) Population	Multiple relevant descriptors	One relevant descriptor	One general descriptor
1b) Intervention	Specific intervention named	General intervention named	Describes properties of intervention, but not helpful keywords
1c) Comparison	Specific alternative named	General intervention named	Describes properties of intervention, but not helpful keywords
1d) Outcome	Specific and objective outcome is named	General outcome named	Vague and not helpful keywords
2) Identify sources to answer the question	At least 4 of: Databases, Journals, Textbooks, Cochrane reviews, EBM publications, Clinical guidelines, Professional organizations	3 sources listed	2 sources listed
3) Search strategy	At least 3 terms related to PICO elements	General description of search strategy	Multiple, non-limited strategies given

Translating Findings Clinically & Refining Clinical Judgment

Only after critically appraising research can the findings be applied in clinical practice by assessing their relevance to specific populations and different clinical presentations. Without this individualized consideration, the application of even the most robust evidence can fall short and lead to suboptimal care or harm. A clear demonstration of the need for thoughtful application of evidence is found in continuing treatment disparities between different patient groups or differing clinical presentations. One retrospective analysis of over 450,000 patients with STEMI and NSTEMI from 2005 to 2015 showed that, despite overall improvement for both groups over time, disparities persist between initial intervention for men and women. Such disparities highlight the need for regular self-assessment and reflection that can help detect factors that influence decision-making and lead to shortcomings.

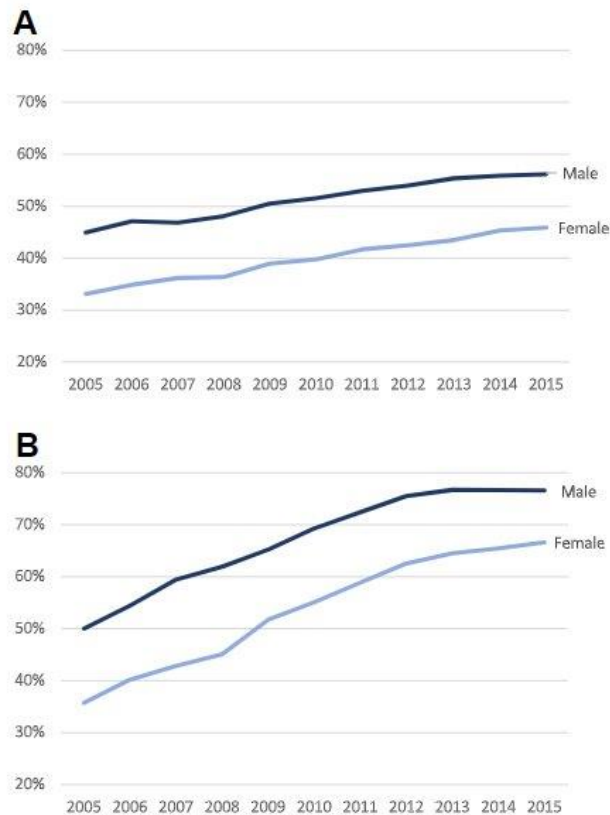


Figure 1. Probability of timely angiography during index admission.
A, NSTEMI according to sex. B, STEMI according to sex.

From Montoy *et al.*, 2022

To improve clinical decision-making, there are at least four key factors that can be addressed. First, medical professionals should actively search for disconfirming evidence, rather than simply accumulating information that supports their initial impressions or hypotheses, to help counteract cognitive biases and promote more objective evaluations. Second, the use of evidence-based algorithms or checklists can serve as cognitive tools to enhance accuracy and reduce reliance on fallible memory. Third, there should be reliance on higher-level evidence or integrative models that synthesize robust research findings, rather than depending solely on anecdotal experience or lower-level evidence, to guide clinical decisions. Fourth, consider the clinical context, as patient-specific factors and situational variables play a significant role in determining the most appropriate course of action. In this way, contextual awareness allows for appropriate application of guidelines while leaving space for clinical judgment. These factors integrated with self-directed life-long learning, critical appraisal of information, and reflective practice can allow you to provide patient-centered care that is both evidence-based and adaptable to the complexities of modern clinical practice.

Further Reading:

- 2024. “Scientific publishers are producing more papers than ever.” *The Economist*.
- 2022. Montoy JCC, Shen YC, Hsia RY. “Trends in Inequities in the Treatment of and Outcomes for Women and Minorities with Myocardial Infarction.” *Ann Emerg Med*.
- 2015. Flacco ME, Manzoli L, Boccia S, Capasso L, Aleksovska K, Rosso A, Scaioli G, De Vito C, Siliquini R, Villari P, Ioannidis JP. “Head-to-head randomized trials are mostly industry sponsored and almost always favor the industry sponsor”. *J Clin Epidemiol*.
- 2014. Khabsa M, Giles CL. “The number of scholarly documents on the public web.” *PLoS One*.
- 2010. Jinha AE. “Article 50 million: an estimate of the number of scholarly articles in existence”. *Learned Publishing*
- 2011. Densen P. “Challenges and opportunities facing medical education.” *Trans Am Clin Climatol Assoc*.
- 2003. Ramos KD, Schafer S, Tracz SM. “Validation of the Fresno test of competence in evidence based medicine.” *BMJ*.
- 1987. Corey GA, Merenstein JH. “Applying the Acute Ischemic Heart Disease Predictive Instrument”. *J Fam Pract*.